

Peripherals: Working Document

Digital Audio Player • Rev 4

Author: Alexander Loker. Design decisions, component selection, analysis and verification.

Drafted by: Claude (Anthropic). Structure and prose from the author's direction and source data.

Contents

1	Summary	1
2	Design Requirements	1
3	Interface	1
4	External Components	1
5	Open Problems	2
6	Revision Log	2
7	Sources	2

1. Summary

This document covers the user-interface peripherals: the display, the front navigation inputs, and the side controls. All inputs connect to ESP32-S3 GPIOs; the display connects over SPI.

Peripheral	Qty	Part / Status
Display	1	BuyDisplay ER-TFT024IPS-3: 2.4 in IPS, 240 × 320, ST7789V, 4-wire SPI, no touch
Front switches (up, down, select)	3	C&K PTS810 SJM 250 SMTR LFS, top-actuated SMD
Rotary encoder	1	Alps EC10E1220501, 10 mm hollow-shaft incremental encoder
Side switches (power, volume up, volume down)	3	E-Switch TL1014BF220QG, side-actuated SMD

2. Design Requirements

Requirement	Value
Display	2.4 in IPS, 240 × 320, SPI; ST7789 or ILI9341 driver; glass within ≈ 43 × 60 mm (inside the battery footprint); vendor must publish the FPC pinout and backlight configuration
Front navigation	Three switches (up, down, select) and one rotary encoder on the front face
Side controls	Power, volume up and volume down on the enclosure side
Wake from standby	Any input that wakes the player must use an RTC-capable ESP32-S3 GPIO
Enclosure	≈ 56 × 75 × 13.2 mm; switch and encoder heights must fit
Constraints	Low noise · Small footprint

3. Interface

Peripheral	Connection
Front and side switches	One ESP32-S3 GPIO each, pulled up, switch to GND; debounced in firmware or with an RC filter
Rotary encoder	Mechanical contacts A and B to two ESP32-S3 GPIOs decoded by the pulse counter (PCNT) peripheral, each with a pull-up and an RC noise filter; common terminal to GND. No supply current, so it can act as a wake source.
Display	SPI (SCLK, MOSI, CS, D/C), reset, and a backlight control output

4. External Components

Ref	Part	Notes	Qty
SW_UP, SW_DN, SW_SEL	C&K PTS810 SJM 250 SMTR LFS	Top-actuated, J-lead SMD; 4.2×3.2 mm; 2.5 mm actuator height; 160 gf; 0.15 mm travel; 150,000 cycles; 50 mA at 16 V DC; -40 to 85 °C. The enclosure key presses the actuator.	3
ENC	Alps EC10E1220501	Incremental (A and B), 10 mm hollow shaft, 9.0 mm mounted height, 24 detents, 12 pulses per revolution, 100,000 cycles, rated 1 mA at 5 V DC. The knob is a custom part through the hollow shaft.	1
SW_PWR, SW_VOLU, SW_VOLD	E-Switch TL1014BF220QG	Side-actuated, right-angle SMD; 4.7×3.5 mm; 220 gf; 50 mA at 12 V DC. The enclosure button presses the actuator.	3
LCD	BuyDisplay ER-TFT024IPS-3 (no-touch version)	2.4 in IPS TFT, 240×320 , ST7789V driver. Supports 8080 parallel (8/9/16/18-bit), 3-wire and 4-wire SPI; used in 4-wire SPI. FPC tail designed for a ZIF connector. Unused parallel data pins (DB0 to DB17) are terminated per the datasheet.	1

5. Open Problems

#	Problem
1	Display details from the ER-TFT024IPS-3 datasheet: backlight configuration (parallel LEDs need no boost converter; a series string does), outline and FPC exit direction, IM0 to IM3 settings for 4-wire SPI, termination of unused pins, and the mating ZIF connector part.
2	Encoder pull-up and RC filter values, from the EC10E catalog.
3	GPIO assignment, including RTC-capable pins for every wake source.

6. Revision Log

Rev	Date	Change
0	2026-09-27	Initial issue: peripheral set, requirements, interface.
1	2026-09-27	Side switches: E-Switch TL1014BF220QG.
2	2026-09-27	TS126CJ specifications recorded; encoder changed to Grayhill 62V11-02-P.
3	2026-09-27	Front switches changed to three C&K PTS810 (up, down, select); encoder changed to Alps EC10E1220501.
4	2026-09-27	Display selected: BuyDisplay ER-TFT024IPS-3 (ST7789V, 4-wire SPI).

7. Sources

-
1. [ER-TFT024IPS-3 Product Page and Datasheet \(BuyDisplay / EastRising\)](#)
 2. [PTS810 SJM 250 SMTR LFS Product Page and Datasheet \(DigiKey\)](#)
 3. [EC10E Series Product Page and Catalog \(Alps Alpine\)](#)
 4. [EC10E1220501 Product Page \(DigiKey\)](#)
 5. [TL1014BF220QG Product Page and Datasheet \(DigiKey\)](#)